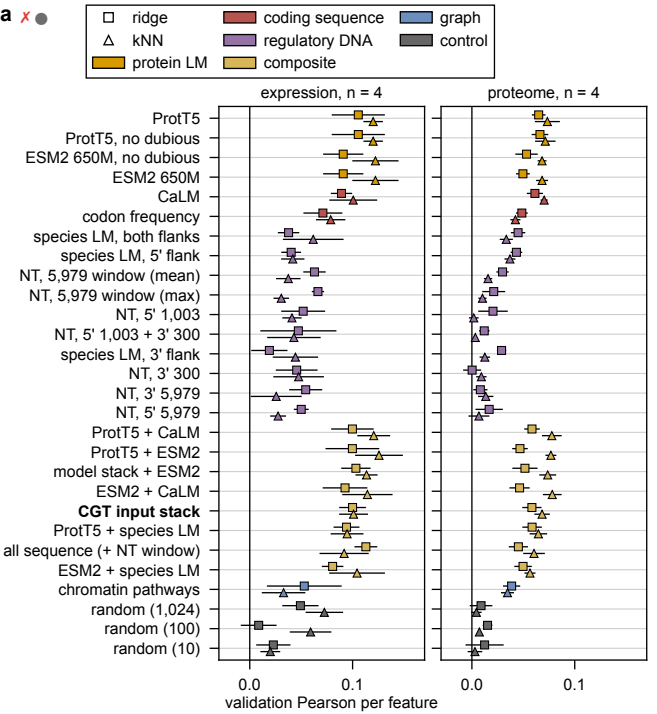
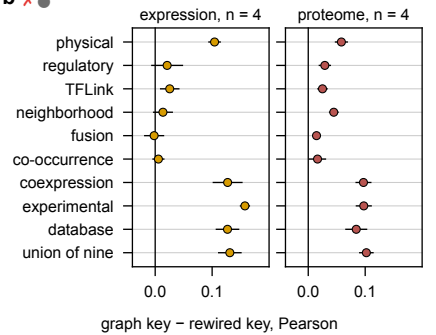
section status ✗ not reviewed, nothing here is approved *figure status* ● on file, in a committed result file ◐ partial, unfinished or unscored ○ idea only, nothing run

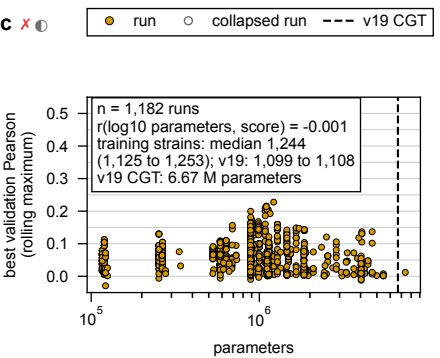
a ✗ ●



b ✗ ●



c ✗ ◐



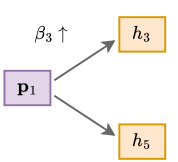
d ✗ ◐ **As drawn in Fig 1f**

$$\beta_{i,t} = \sigma\left(\frac{(h_i W_Q)\cdot(\mathbf{p}_t W_K)}{\sqrt{d_k}}\right)$$

$$\Delta h_i = \sum_t \beta_{i,t}\,(\mathbf{p}_t W_V)$$

$$h_i^{\mathrm{p}} = h_i + \Delta h_i$$

$$p = \{(e_t, \tau_t, m_t)\}_{t=1}^{M}$$



With one perturbed gene, weights that differ between genes need the function to act on each pair, a sigmoid. Methods write a softmax over t.

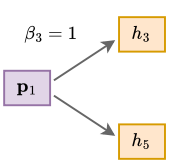
As run in v19 and v20 (null sink off)

$$\beta_{i,t} = \underset{t\in[M]}{\mathrm{softmax}}\,\frac{(h_i W_Q)\cdot(\mathbf{p}_t W_K)}{\sqrt{d_k}}$$

$$\Delta h_i = \sum_t \beta_{i,t}\,(\mathbf{p}_t W_V)$$

$$h_i^{\mathrm{p}} = h_i + \Delta h_i$$

$$p = \{(e_t, \tau_t, m_t)\}_{t=1}^{M}$$



Softmax over the perturbed genes: with one perturbed gene the weight is 1 for every gene and the update is the same vector for every gene.